Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Section 1: Domains and Ranges of Inverse Trig Functions

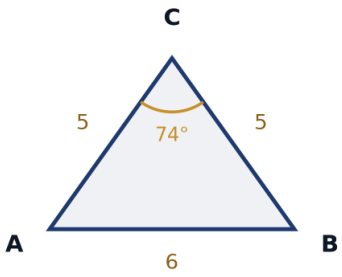
1 What are the domain and range of the principal arcsine function $y = \arcsin x$?

a. Domain: $[-1, 1]$; Range: $[-\frac{\pi}{2}, \frac{\pi}{2}]$

b. Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

c. Domain: $[-1, 1]$; Range: $[0, \pi]$

d. Domain: $(-\infty, \infty)$; Range: $(-\frac{\pi}{2}, \frac{\pi}{2})$



2 What are the domain and range of $y = \arccos x$?

a. Domain: $[-1, 1]$; Range: $[0, \pi]$

b. Domain: $[-1, 1]$; Range: $[-\frac{\pi}{2}, \frac{\pi}{2}]$

c. Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

d. Domain: $[0, 1]$; Range: $[0, \pi/2]$

3 What are the horizontal asymptotes of the graph of $y = \arctan x$?

a. $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$

b. $y = \pi$ and $y = 0$

c. $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$

d. $y = 1$ and $y = -1$

Section 2: Derivatives of Inverse Trigonometric Functions

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4 What is the derivative of $f(x) = \arcsin(3x)$ for $|x| < \frac{1}{3}$?

a. $\frac{3}{\sqrt{1-9x^2}}$

b. $\frac{1}{\sqrt{1-9x^2}}$

c. $\frac{3}{\sqrt{1-3x^2}}$

d. $\frac{3}{1+9x^2}$

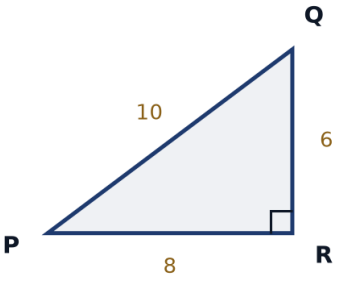
5 Find the derivative of $g(x) = \arctan\left(\frac{x}{2}\right)$.

a. $\frac{2}{4+x^2}$

b. $\frac{1}{1+x^2/4}$

c. $\frac{1}{4+x^2}$

d. $\frac{2}{\sqrt{4-x^2}}$



6 What is the relationship between the derivatives of $\arcsin x$ and $\arccos x$?

a. $\frac{d}{dx}[\arccos x] = -\frac{d}{dx}[\arcsin x] = -\frac{1}{\sqrt{1-x^2}}$

b. $\frac{d}{dx}[\arccos x] = \frac{d}{dx}[\arcsin x]$

c. $\frac{d}{dx}[\arccos x] = \frac{1}{1-x^2}$

d. They are reciprocals of each other

Section 3: Tangent Lines and Extreme Values

7 Find the slope of the tangent line to $y = \arcsin x$ at $x = 0$.

a. 1

b. 0

c. $\pi/2$

d. Undefined

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8 What happens to the slope of $y = \arcsin x$ as $x \to 1^-$?

a. The slope approaches $+\infty$ (vertical tangent)

b. The slope approaches 0

c. The slope approaches 1

d. The slope approaches -1

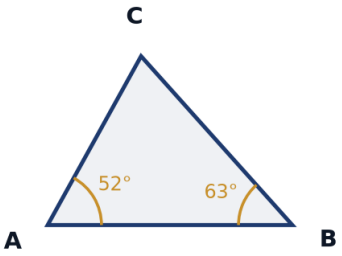
9 Find the equation of the tangent line to $y = \arctan x$ at $x = 1$.

a. $y - \frac{\pi}{4} = \frac{1}{2}(x - 1)$

b. $y - \frac{\pi}{4} = 1(x - 1)$

c. $y = \frac{1}{2}x + \frac{\pi}{4}$

d. $y - 1 = \frac{1}{2}(x - \frac{\pi}{4})$



Section 4: Integration Producing Inverse Trig Functions

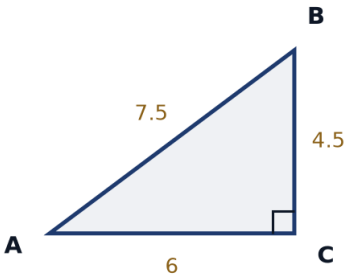
10 Evaluate $\int_0^{1/2} \frac{1}{\sqrt{1 - x^2}} dx$.

a. $\frac{\pi}{6}$

b. $\frac{\pi}{3}$

c. $\frac{\pi}{4}$

d. $\frac{1}{2}$



11 Evaluate $\int_0^1 \frac{1}{1 + x^2} dx$.

a. $\frac{\pi}{4}$

b. $\frac{\pi}{2}$

c. 1

d. $\ln 2$

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12 Evaluate $\int \frac{1}{\sqrt{4-9x^2}} dx$.

a. $\frac{1}{3}\arcsin\left(\frac{3x}{2}\right) + C$

b. $\frac{1}{2}\arcsin\left(\frac{3x}{2}\right) + C$

c. $\arcsin\left(\frac{3x}{2}\right) + C$

d. $\frac{1}{6}\arcsin\left(\frac{3x}{2}\right) + C$

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